Alexander **Filak**

About me

I am a highly motivated first-year Master's student at the Moscow Institute of Physics and Technology (MIPT), specializing in Artificial Intelligence at the Center of Information Technologies and Systems (CITIS). I am proud to be part of the Phystech School of Radio Engineering and Computer Technology (DREC), which provides me with a strong foundation in cutting-edge technical education and research opportunities.

personal

Alexander Filak
23 years

Areas of specialization

CV • ML

Interests

Computer Science
Systems Programming
Theoretical Physics
Mathematical Physics

@n0t_KNOW
filaka771

Education

2020–2025	**MIPT** Bachelor · DREC
2025–Now	**MIPT** Master · DREC

Experience

2022 – Now **profi.ru/profile/FilakAY**
Tutor in higher mathematics and theoretical physics. Worked with students from MIPT, HSE, MAI, etc.

2023 – Now
Mathematics teacher at the Summer Multidisciplinary School. Author of courses on Group Theory, Differential Equations, etc.

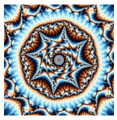
Programming

- C
- C++
- LaTeX
- python
- ASM

Projects

Paracl **github.com/filaka771/Paracl**
Compiler for a simple C-like language in C++, with a hand-written lexer, recursive-descent parser, x86-64 code generation, and a small syscall-based runtime for integer I/O, targeting FASM assembly.

Mandelbrot Set **github.com/filaka771/Mandelbrot-set**
Fractal renderer in C++ featuring multiple Mandelbrot implementations: a scalar CPU version, an AVX2-optimized renderer, a Boost.Multiprecision/MPFR deep-zoom path, and custom PPM image generation with configurable shader-based coloring.



CPU **github.com/filaka771/Cpu**
Virtual CPU in C featuring a full toolchain with a two-pass assembler, binary disassembler, and register/stack-based emulator supporting arithmetic, branching, function calls and indirect addressing.

LRU Cache **github.com/filaka771/LRU**
Generic LRU cache implementation in C++ designed for efficient recency tracking and eviction, with libFuzzer-based randomized testing against a naive vector-based implementation.

LLVM Contribution **github.com/llvm/llvm-project/pull/190570**
Merged upstream LLVM/Clang contribution fixing alignment in `-analyzer-help` output via `AnalyzerOptions::printFormattedEntry`, with added unit tests covering formatting edge cases in analyzer option listing.

Languages

Russian	C2	mother tongue
English	B2	●●●○

Publications

2025 Filak, A. Yu., & Mozolina, N. V. (2025). Study of the applicability of general-purpose hash functions in the problem of data integrity monitoring and deduplication. *Voprosy zashchity informatsii [Information Security Issues]*, 2025(1). EDN: HZQJSI. DOI: 10.52190/2073-2600_2025_1_8.

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